AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 9 · STUDENT LESSON

Experimental units and independence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Baselines and fair comparisons. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of experimental units and independence. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **experimental unit**s and independence.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **experimental unit**s and independence

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Baselines and fair comparisons. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **experimental unit**s and independence using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Experimental units and independence case. Separate observations, given facts and assumptions before choosing an explanation.

The **experimental unit** is the independently assigned or sampled entity. Multiple measurements from one unit are often related.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Experimental units and independence. Connect each labelled input, deciding step and result.

If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.

Experimental units and independence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Three queries each run four times produce12 outputs.	If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.	There are three query units and12 run measurements. Do not analyse them as12 independent queries.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Experimental units and independence. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Three queries each run four times produce12 **output**s. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Put one P1 view into evaluation while other P1 views train.

> **Predicted result**
>
> **Prediction**: Flag object-level leakage for this new-plant target.

> **Why the result follows**
>
> Reason: Evaluation now includes a plant already represented during **training**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Experimental units and independence to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Experimental unit	The entity independently sampled or assigned
Repeated measure	Another observation on the same unit
Clustering	Related measurements grouped within an entity

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Experimental units and independence, predict a step, then compare with its explanation.

Three queries each run four times produce12 **output**s.

There are three query units and12 run measurements. Do not analyse them as12 independent queries.

Worked explanation · Experimental units and independence

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

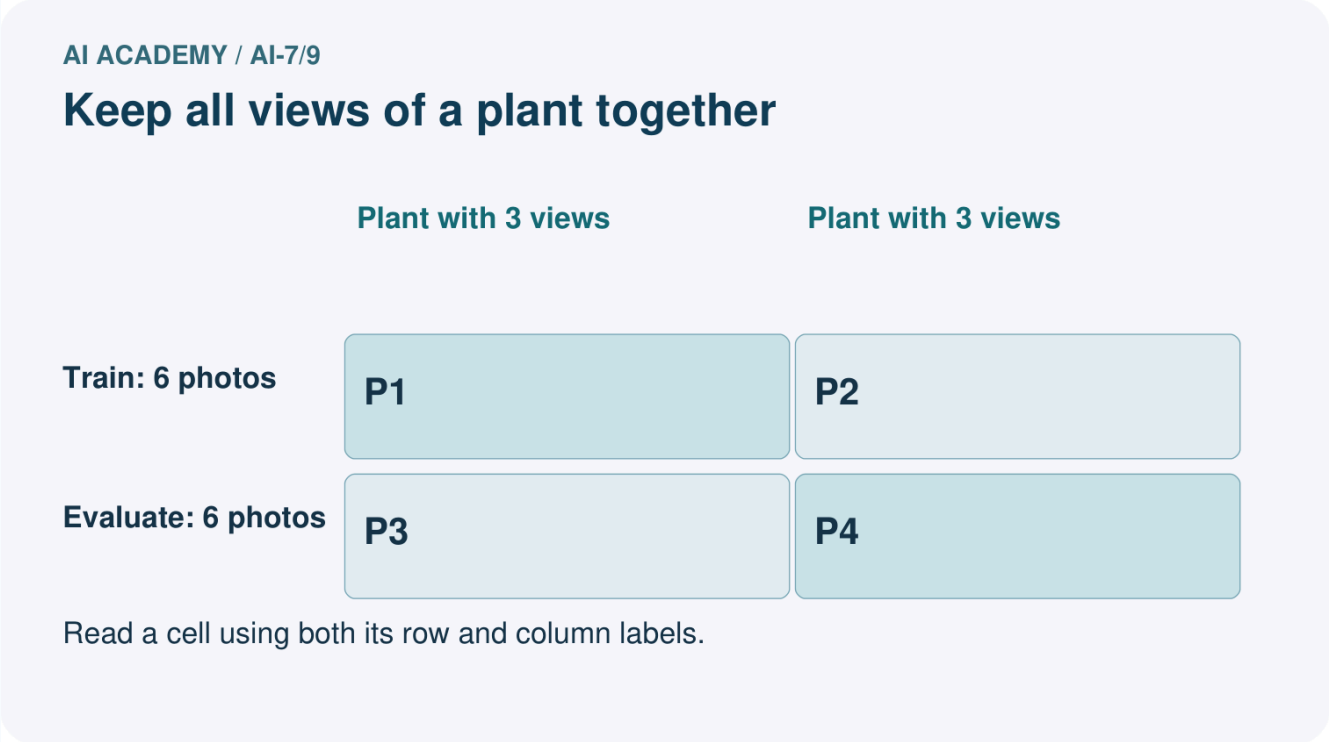
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Experimental units and independence

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

The facts we will use

> **Supplied facts**
>
> Plant IDs P1–P4 each have three views. All views of a plant must stay in one split.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/9

Experimental units and independence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the unit

Four plants, not twelve independent plants, generated the observations.



2. Group before splitting

Assign whole plant IDs to training or evaluation.



3. Trace a valid split

Train on P1,P2 and evaluate on P3,P4: six photos per split.

CHANGE ONE THING

Put one P1 view into evaluation while other P1 views train.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the unit	Four plants, not twelve independent plants, generated the observations.
2. Group before splitting	Assign whole plant IDs to training or evaluation.
3. Trace a valid split	Train on P1,P2 and evaluate on P3,P4: six photos per split.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Repeated views from one object share information. Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

What this example does not establish

Keep the stated limit

Four plants offer little population coverage; even a group-correct split may be too small for broad conclusions.

Keep the stated limit

Read the labelled picture: Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

Keep the stated limit

Change one thing: Put one P1 view into evaluation while other P1 views train. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Experimental units and independence claim and explain a repair using the supplied evidence.

Independence follows design and relationships, not row count.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Independence follows design and relationships, not row count. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Experimental units and independence with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Experimental unit**s and independence". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Experimental unit**s and independence. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Experimental units and independence. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Three queries each run four times produce12 **output**s. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	

Record	Your evidence
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Experimental units and independence. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Independence follows design and relationships, not row count. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **experimental unit**s and independence **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **experimental unit**s and independence.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Experimental units and independence case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Experimental units and independence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Three queries each run four times produce12 outputs.	If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.	There are three query units and12 run measurements. Do not analyse them as12 independent queries.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Five classrooms each contribute20 student responses under a classroom-level assignment. Identify assignment units and measurements.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Five classrooms each contribute20 student responses under a classroom-level assignment. Identify assignment units and measurements.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Experimental units and independence, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

By the end, I can	How I will show it
Explain the mechanism and terminology of experimental units and independence	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Experimental units and independence without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **experimental unit**s and independence

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Train on P1,P2 and evaluate on P3,P4: six photos per split.

1. Group before splitting

2. Identify the unit

3. Trace a valid split

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Trace a valid split Train on P1,P2 and evaluate on P3,P4: six photos per split. Repeated views from one object share information.

Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Put one P1 view into evaluation while other P1 views train. Which conclusion is supported by the supplied facts?

1. Flag object-level leakage for this new-plant target.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Flag object-level leakage for this new-plant target. Evaluation now includes a plant already represented during **training**.

Repair a missing building block

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

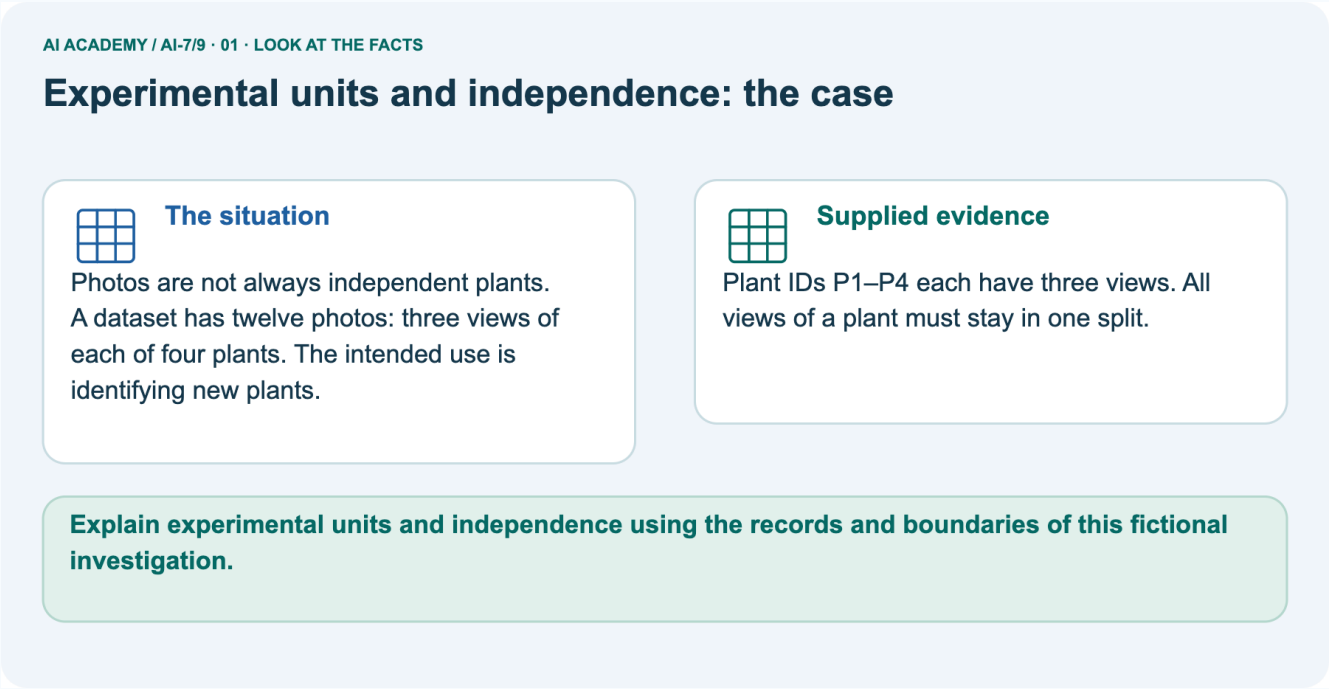
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Experimental units and independence graphic using its labels, deciding evidence and relevant limitation.

01. Experimental units and independence: the case



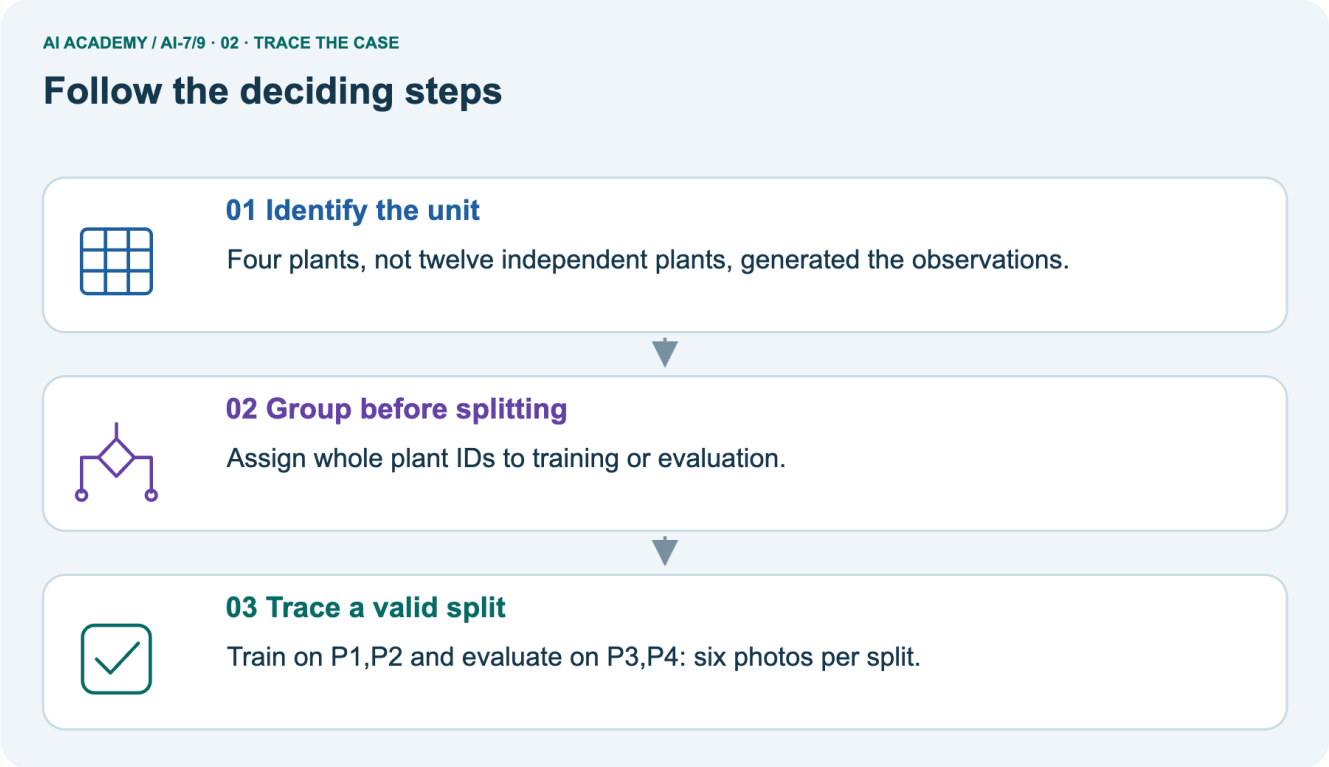
Follow the labelled picture with the matching explanation. Open the vector diagram

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants. Supplied facts: Plant IDs P1–P4 each have three views.

All views of a plant must stay in one split.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

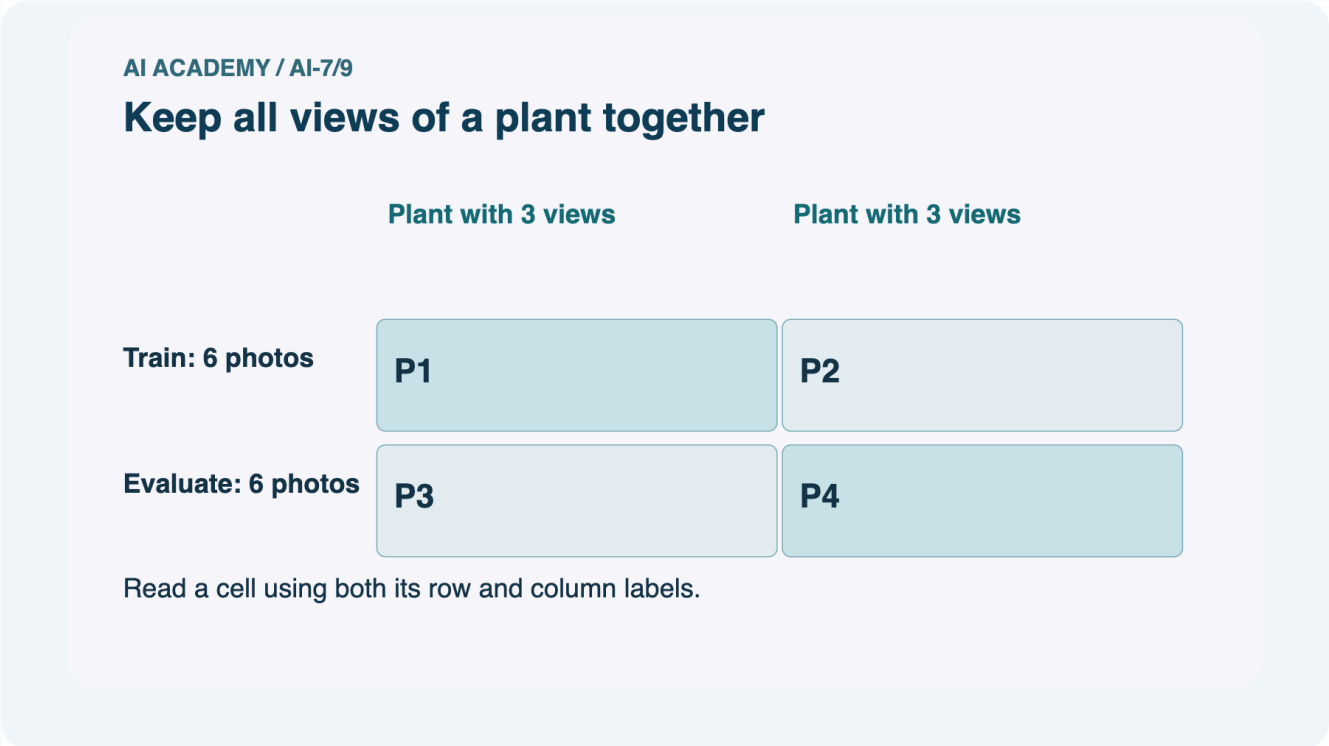


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the unit: Four plants, not twelve independent plants, generated the **observation**s. → Group before splitting: Assign whole plant IDs to **training** or evaluation. → Trace a valid split: Train on P1,P2 and evaluate on P3,P4: six photos per split.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep all views of a plant together

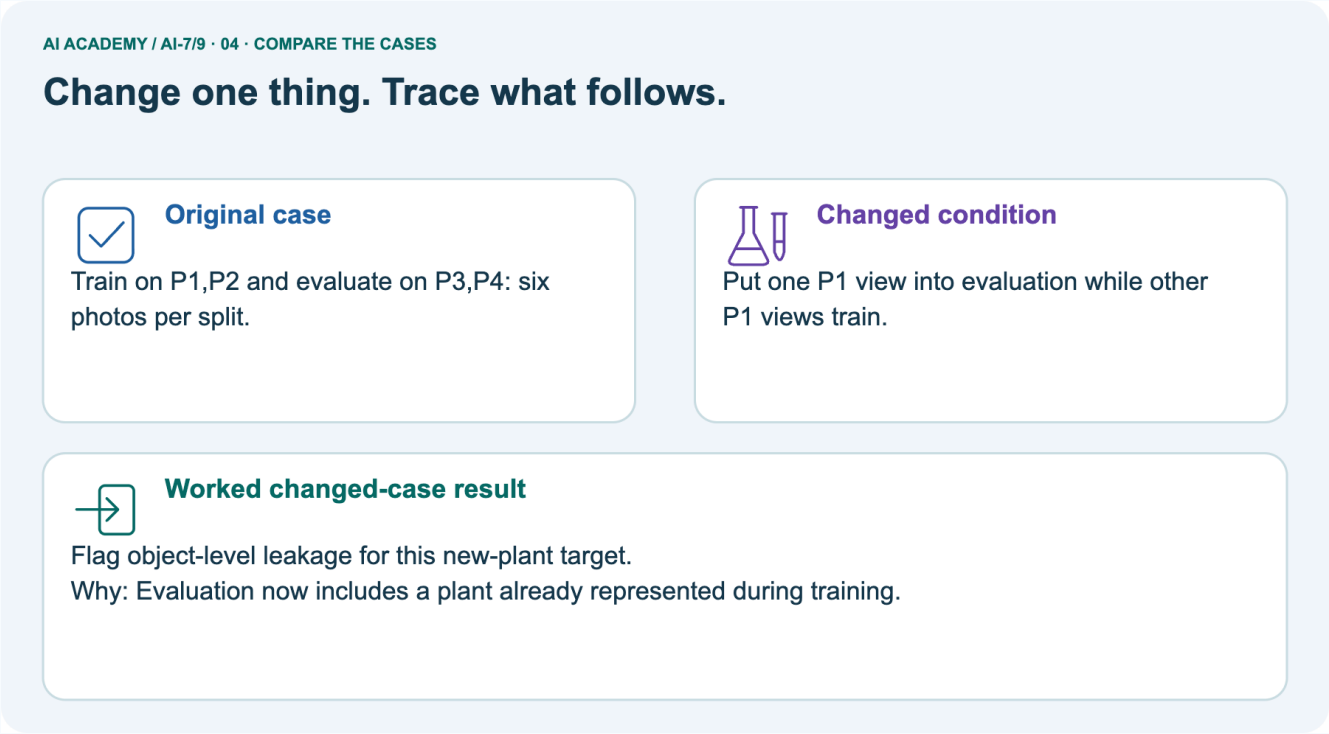


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep all views of a plant together. Train: 6 photos: Plant with 3 views = P1, Plant with 3 views = P2. Evaluate: 6 photos: Plant with 3 views = P3, Plant with 3 views = P4.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



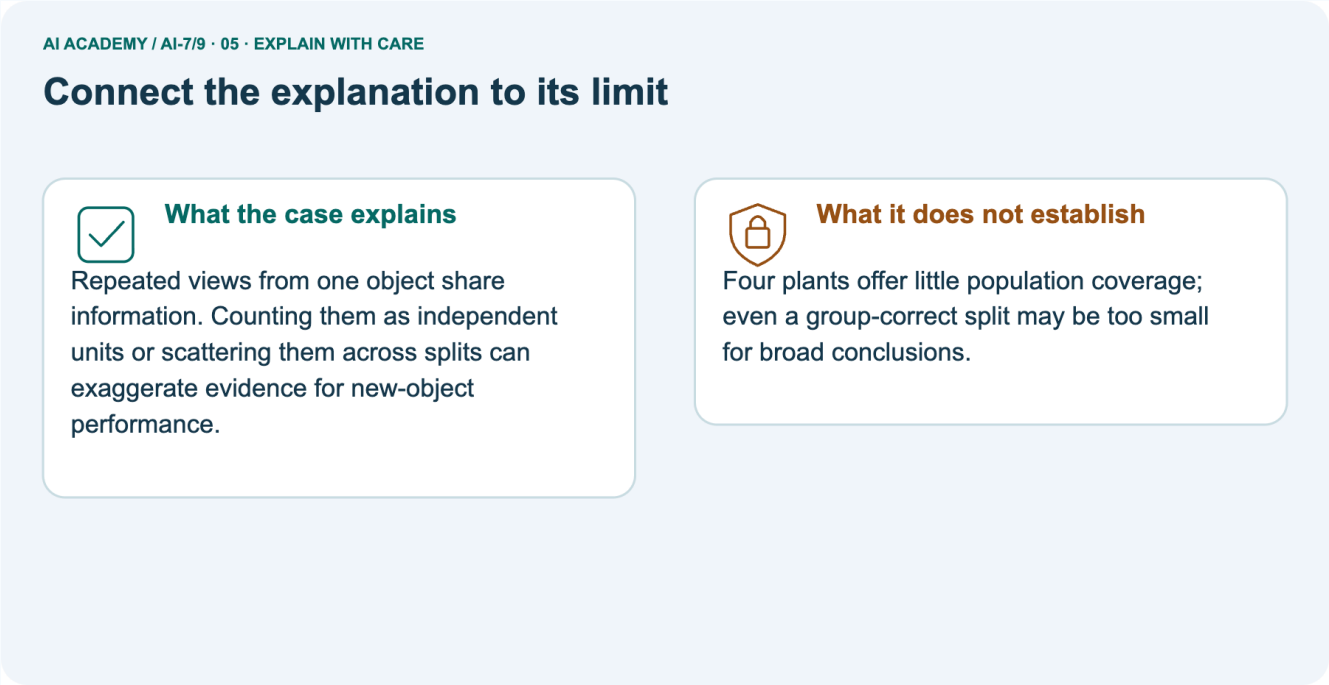
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Train on P1,P2 and evaluate on P3,P4: six photos per split. Changed condition: Put one P1 view into evaluation while other P1 views train. Expected result: Flag object-level leakage for this new-plant target.

Reason: Evaluation now includes a plant already represented during **training**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



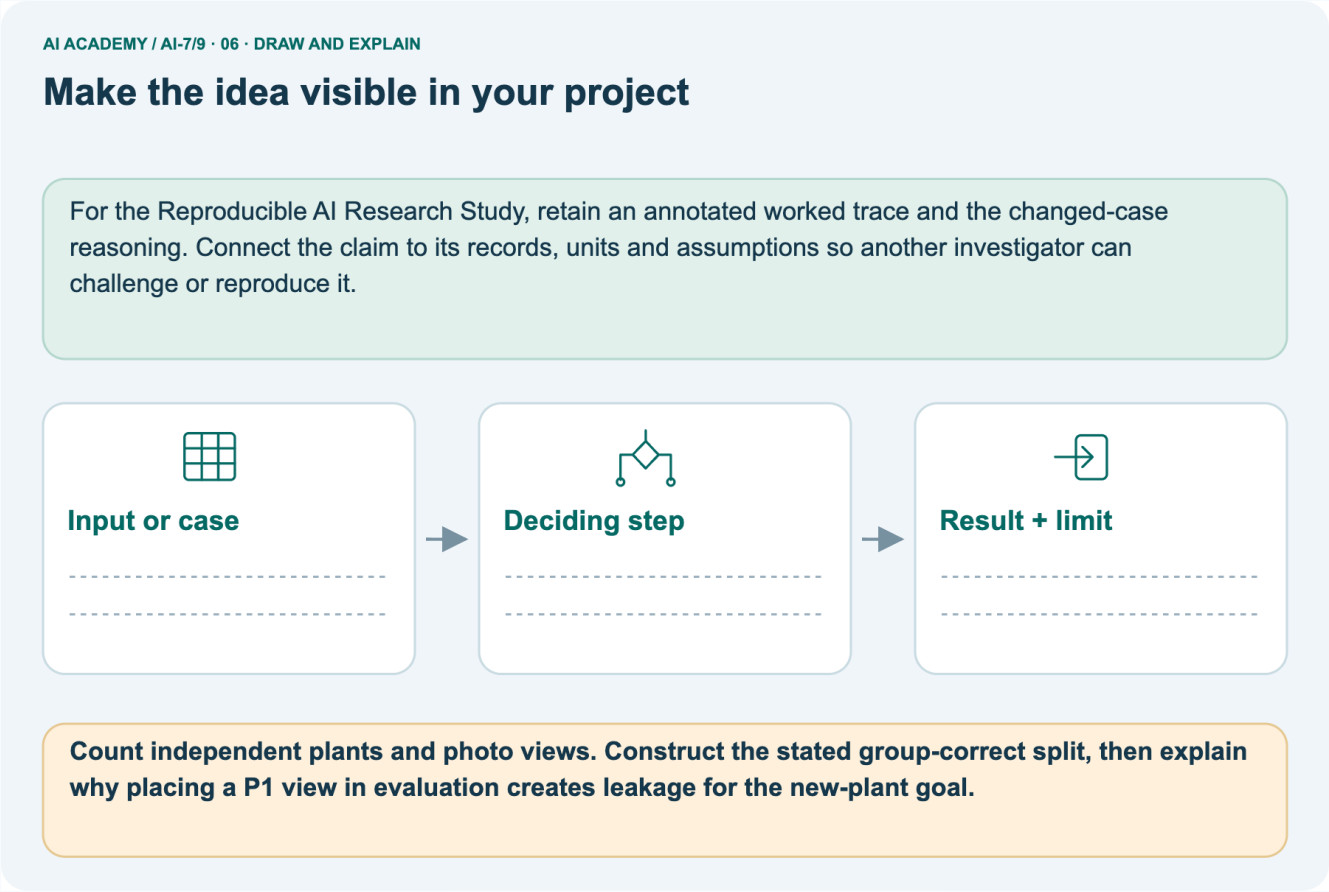
Follow the labelled picture with the matching explanation. Open the vector diagram

Repeated views from one object share information. Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

Limit: Four plants offer little population coverage; even a group-correct split may be too small for broad conclusions.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.